KON414E - Principles of Robot Autonomy
2025-2026 Fall Term
Homework 3 **Deadline: 14 December, 23:59**

Homework 3: World Integration and Robot Operation

Important Note: Document **each step** with screenshots and brief explanations.
You are expected to complete this homework **individually**, not as a team.

Use your existing robot model from *Homework 1* and sensor configuration from *Homework 2*.

Objective: To load the provided world environment into your ROS/Gazebo setup, spawn your sensor-integrated robot inside this world, demonstrate robot movement, visualize all assigned sensor outputs in RViz, and prepare a short report and demonstration video.

1. Prepare a single launch file that runs all required nodes (e.g., Gazebo, RViz, robot spawning, controller, etc.) to perform the following tasks. Add your launch file to your report.
   - Import the world file provided under the Class Files section on Ninova into your Gazebo Sim or Gazebo Classic environment. Apply any required adjustments (e.g., plugin or SDF fixes) and demonstrate that the world runs correctly on your Gazebo version.
     If you experience issues adapting the provided world file to your Gazebo version, contact the professor or the assistants.
   - Launch your robot inside this world. Verify that the robot spawns properly and that all sensors are active.
   - Run the ROS node you developed in Homework 1 to move the robot. Demonstrate robot motion inside the provided environment (e.g., moving through corridors or around obstacles).
   - Visualize your robot and all assigned sensor outputs in RViz. Perform the following:
     - Camera and LiDAR / 3D LiDAR: Display the data in RViz and include this visualization in your **video and report**.
     - For sensors such as IMU or GPS, plot relevant data streams using `rqt_multiplot`, PlotJuggler, or an equivalent tool. Include these plots in your **video and report**.
2. Record a video showcasing your system in action (displaying both RViz and Gazebo). Make sure your video demonstrates the robot moving inside the world and all assigned sensors functioning. Upload the video to a shareable platform (e.g., Google Drive) and ensure the video is accessible to anyone with the link.
3. Document each step in a report, including screenshots and brief explanations of your actions. Include the video link in your report.